

1. let $B = X^{-1} + A$ Thus $dB = d(X^{-1}) = -X^{-1}(dX)X^{-1}$

we have: $d(\det B) = \det B \cdot \text{Tr}(B^{-1}dB)$

Therefore $df = \det B \cdot \text{Tr}(B^{-1}(-X^{-1}(dX)X^{-1}))$
 $= -\det B \cdot \text{Tr}(B^{-1}X^{-1}(dX)X^{-1})$
 $= -\det B \cdot \text{Tr}(X^{-1}B^{-1}X^{-1}dX)$

from the definition of gradient,

~~$\nabla_x f$~~ $df = \text{Tr}((\nabla_x f)^T dX)$

we have: $\text{Tr}((\nabla_x f)^T dX) = -\det B \cdot \text{Tr}(X^{-1}B^{-1}X^{-1}dX)$

which leads to: $(\nabla_x f)^T = -\det B \cdot (X^{-1}B^{-1}X^{-1})$

Thus: $\nabla_x f = -\det B \cdot (X^{-1}B^{-1}X^{-1})^T$
 $= -\det B \cdot (X^{-1})^T (B^{-1})^T (X^{-1})^T$
 $= -\det(X^{-1} + A) \cdot (X^{-1})^T (X^{-1} + A)^{-1} (X^{-1})^T$

2. (a) Let $M(t) = A + tI_n$, because $A \in S_+^n$ and $t > 0$

Thus $M(t)$ is positive, which is $M(t) \in S_{++}^n$

And that $M(t)$ is non-degenerate, $M^{-1}(t)$ exists.

Let $y(t) = M^{-1}(t)b$, then $f(t) = \|y(t)\|$

$$\frac{dM(t)}{dt} = I_n \quad \frac{dM^{-1}(t)}{dt} = -M^{-1}(t) \left(\frac{dM}{dt} \right) M^{-1}(t)$$

$$\text{Therefore } \frac{dy}{dt} = \frac{d}{dt}(M^{-1}b) = \frac{dM^{-1}}{dt}b = -M^{-1}I_n M^{-1}b = -M^{-2}b$$

$$\text{Apparently, } f(t) = \|y(t)\| = \sqrt{y(t)^T y(t)}$$

$$\text{Then, } f'(t) = \frac{1}{2}(y^T y)^{-\frac{1}{2}} \cdot 2 y^T \frac{dy}{dt} = \frac{y^T \frac{dy}{dt}}{\|y\|}$$

$$\text{we know } \frac{dy}{dt} = -M^{-2}b \text{ so,}$$

$$f'(t) = \frac{y^T(-M^{-2}b)}{\|y\|} = -\frac{y^T M^{-2}b}{\|y\|}$$

$$\text{and } y = M^{-1}b, \text{ so that } y^T = b^T M^{-1}$$

$$\Rightarrow f'(t) = -\frac{1}{\|y(t)\|} b^T M^{-3}b$$

$$\text{Thus } \nabla f(t) = f'(t) = -\frac{b^T (A + tI_n)^{-3}b}{\|(A + tI_n)^{-1}b\|}$$

$$\text{let } u(t) = -b^T M^{-3}b \quad v(t) = \|y(t)\| = \|M^{-1}b\|$$

$$u'(t) = -\frac{d}{dt}(b^T M^{-3}b) = -(-3b^T M^{-4}b) = 3b^T M^{-4}b$$

$$v'(t) = \frac{y^T \frac{dy}{dt}}{\|y\|} = -\frac{y^T M^{-2}b}{\|y\|} = -\frac{b^T M^{-3}b}{\|y\|}$$

$$\text{we have } f''(t) = \frac{u'(t)v(t) - u(t)v'(t)}{[v(t)]^2}$$

$$= \frac{3b^T M^{-4}b \cdot \|y\| - (-b^T M^{-3}b)(-\frac{b^T M^{-3}b}{\|y\|})}{[\|y\|]^2}$$

$$= \frac{3b^T M^{-4}b \cdot [\|y\|]^2 - [b^T M^{-3}b]^2}{[\|y\|]^3}$$

$$\text{Thus } \nabla^2 f(t) = \frac{3b^T M^{-4}b \cdot \|(A + tI_n)^{-1}b\|^2 - [b^T (A + tI_n)^{-3}b]^2}{\|(A + tI_n)^{-1}b\|^3}$$

Q 2.(b) We know the Frobenius Norm:

for a matrix $M \in \mathbb{R}^{n \times n}$,

$$\|M\|_F = \sqrt{\text{Tr}(M^T M)} = \sqrt{\sum_{i,j} M_{ij}^2}$$

Thus: $\|M\|_F^2 = \text{Tr}(M^T M)$

And $f(x)$ can be written as: $f(x) = \frac{1}{2} \text{Tr}((xx^T - A)^T (xx^T - A))$

xx^T and A are symmetric, their ~~transposes~~ transposes are themselves:

$$f(x) = \frac{1}{2} \text{Tr}((xx^T - A)(xx^T - A)) \\ = \frac{1}{2} \text{Tr}(xx^T xx^T - xx^T A - Axx^T + A^2)$$

$$\text{Tr}(xx^T xx^T) = \text{Tr}(x(x^T x)x^T) = \text{Tr}(x^T x)^2$$

$$\text{Tr}(xx^T A) = \text{Tr}(Axx^T) = x^T Ax$$

$$\text{Tr}(Axx^T) = x^T Ax$$

$\text{Tr}(A^2)$ is a constant.

$$f(x) = \frac{1}{2} [(x^T x)^2 - 2x^T Ax + \text{Tr}(A^2)]$$

$$= \frac{1}{2} (x^T x)^2 - x^T Ax + \frac{1}{2} \text{Tr}(A^2)$$

Let $s = x^T x$, then $\frac{1}{2} (x^T x) = \frac{1}{2} s^2$

$$\nabla \left(\frac{1}{2} s^2 \right) = s \nabla s = s \cdot 2x = 2(x^T x)x$$

$$\nabla (-x^T Ax) = -(A + A^T)x = -2Ax \quad (\text{since } A \text{ is symmetric})$$

Therefore: $\nabla f(x) = 2(x^T x)x - 2Ax$.

$$\nabla^2 f(x) = \nabla (\nabla f(x))$$

$$\nabla (2(x^T x)x) = 2(\nabla(x^T x)x + (x^T x)\nabla x)$$

$$= 2(2xx^T + (x^T x)I_n)$$

$$= 4xx^T + 2(x^T x)I_n$$

$$\nabla (-2Ax) = -2A$$

Thus $\nabla^2 f(x) = 4xx^T + 2(x^T x)I_n - 2A$.

$$\begin{aligned}
 3.(a) \quad \nabla f(x) &= \nabla (\langle a, x \rangle - \ln(1 - \langle b, x \rangle)) \\
 &= a - \nabla [\ln(1 - \langle b, x \rangle)] \\
 &= a - \frac{(-b)}{1 - \langle b, x \rangle} \\
 &= a + \frac{b}{1 - \langle b, x \rangle}
 \end{aligned}$$

Let $\nabla f(x) = 0$, then $\frac{b}{1 - \langle b, x \rangle} = -a$
 we know $a, b \neq 0$

Since $\frac{1}{1 - \langle b, x \rangle}$ is a scalar, there must be a scalar c which leads to:

$$a = cb, \quad c \in \mathbb{R}$$

replace a with cb in equation:

$$\frac{b}{1 - \langle b, x \rangle} = -cb \Rightarrow \langle b, x \rangle = 1 + \frac{1}{c}$$

Since $x \in E = \{x \in \mathbb{R}^n \mid \langle b, x \rangle < 1\}$,

$$1 + \frac{1}{c} < 1 \Rightarrow c < 0$$

Thus when $a = cb$ and $c < 0$, we have stationary points which:

$$\langle b, x \rangle = 1 + \frac{1}{c}$$

the stationary point set:

$$\{x \in \mathbb{R}^n \mid \langle b, x \rangle = 1 + \frac{1}{c}, c < 0\}$$

$$3. (b) \quad \nabla f(x) = c \cdot e^{-\langle Ax, x \rangle} + \langle c, x \rangle \cdot e^{-\langle Ax, x \rangle} \cdot (-2Ax)$$

$$= e^{-\langle Ax, x \rangle} [c - 2\langle c, x \rangle Ax]$$

$$\text{let } \nabla f(x) = 0, \text{ since } e^{-\langle Ax, x \rangle} > 0$$

$$c - 2\langle c, x \rangle Ax = 0$$

$$\Rightarrow c = 2\langle c, x \rangle Ax$$

Since $c \neq 0$ let $\lambda = \langle c, x \rangle$. then

$$c = 2\lambda Ax$$

$$\Rightarrow x = \frac{1}{2\lambda} A^{-1}c$$

$$\text{therefore } \lambda = \frac{1}{2\lambda} \langle c, A^{-1}c \rangle, \text{ and } \lambda \neq 0$$

$$\text{then } 2\lambda^2 = \langle c, A^{-1}c \rangle$$

$$\lambda = \pm \sqrt{\frac{1}{2} \langle c, A^{-1}c \rangle}$$

$\langle c, A^{-1}c \rangle > 0$, since A is positive definite

when $\lambda = \sqrt{\frac{1}{2} \langle c, A^{-1}c \rangle}$,

$$x = \frac{1}{2\lambda} A^{-1}c = \frac{A^{-1}c}{2\sqrt{\frac{1}{2} \langle c, A^{-1}c \rangle}}$$

$$= \frac{A^{-1}c}{\sqrt{2 \langle c, A^{-1}c \rangle}}$$

$$\lambda = \langle c, x \rangle \Rightarrow \lambda = \langle c, \frac{A^{-1}c}{\sqrt{2 \langle c, A^{-1}c \rangle}} \rangle = \frac{\sqrt{\langle c, A^{-1}c \rangle}}{\sqrt{2}}$$

similarly, $\lambda = -\sqrt{\frac{1}{2} \langle c, A^{-1}c \rangle}$ exist.

$$x = \frac{1}{2\lambda} A^{-1}c = \pm \frac{A^{-1}c}{\sqrt{2 \langle c, A^{-1}c \rangle}}$$

$$\text{where } \lambda = \pm \frac{\sqrt{\langle c, A^{-1}c \rangle}}{\sqrt{2}}$$

Since $\langle c, A^{-1}c \rangle > 0$, stationary points constantly exist.

3. (c)

$$f(X) = \langle X^{-1}, I_n \rangle - \langle A, X \rangle$$

$$= \text{tr}(X^{-1}) - \text{tr}(AX)$$

$$\text{in which, } \frac{\partial}{\partial X} \text{tr}(X^{-1}) = -X^{-1}X^{-1} = -X^{-2}$$

$$\frac{\partial}{\partial X} \text{tr}(AX) = A.$$

$$\nabla f(X) = \frac{\partial}{\partial X} (\text{tr}(X^{-1}) - \text{tr}(AX))$$

$$= -X^{-2} - A.$$

$$\text{let } \nabla f(X) = 0 \quad \text{then} \quad X^{-2} = -A$$

~~$X^{-2} \in S_{++}$~~ Since $X^* \in S_{++}^n$, X^{-2} is positive definite, Thus $-A$ is positive definite, and A is negative definite.

$$\text{Therefore, } X^{-2} = -A \Rightarrow X = (-A)^{-\frac{1}{2}}$$

Stationary Point: $X = (-A)^{-\frac{1}{2}}$, where A is negative definite.

if A is non-negative definite

Stationary point doesn't exist.

4. Original convolution:

$$Y_{i,j} = \sum_{u,v,w=0}^{h-1,w-1,C-1} K_{u,v,w} X_{i+u,j+v,w} \quad \begin{cases} 0 \leq i \leq H-h \\ 0 \leq j \leq W-w \end{cases}$$

in which the w was used as the kernel width and index for channels.

So to eliminate ambiguity:

$$Y_{ij} = \sum_{u=0}^{h-1} \sum_{v=0}^{w-1} \sum_{c=0}^{C-1} K_{u,v,c} X_{i+u,j+v,c}$$

$$\textcircled{1} \quad \frac{\partial f}{\partial K_{u,v,c}} = \sum_{i=0}^{H-h} \sum_{j=0}^{W-w} \frac{\partial f}{\partial Y_{i,j}} \cdot \frac{\partial Y_{i,j}}{\partial K_{u,v,c}}$$

according to the convolution definition:

$$\frac{\partial Y_{i,j}}{\partial K_{u,v,c}} = X_{i+u,j+v,c}$$

Therefore,

$$\frac{\partial f}{\partial K_{u,v,c}} = \sum_{i=0}^{H-h} \sum_{j=0}^{W-w} \nabla_Y f_{i,j} \cdot X_{i+u,j+v,c}$$

which means, $\nabla_K f$ is the convolution of X and $\nabla_Y f$: $\nabla_K f = X * \nabla_Y f$

② Similarly,

$$\frac{\partial f}{\partial X_{i,j,c}} = \sum_{u=0}^{h-1} \sum_{v=0}^{w-1} \frac{\partial f}{\partial Y_{i-u,j-v}} \cdot \frac{\partial Y_{i-u,j-v}}{\partial X_{i,j,c}}$$

Thus

$$\frac{\partial Y_{i-u,j-v}}{\partial X_{i,j,c}} = K_{u,v,c}$$

Therefore

$$\frac{\partial f}{\partial X_{i,j,c}} = \sum_{u=0}^{h-1} \sum_{v=0}^{w-1} \nabla_Y f_{i-u,j-v} \cdot K_{u,v,c}$$

which means, if we define $K^{\text{rot}180}$ is to rotate K by 180° , then:

$$\nabla_X f = \nabla_Y f * K^{\text{rot}180}$$

Summary:

$$\nabla_K f = X * \nabla_Y f \quad (\text{convolution of Image } X \text{ and } \nabla_Y f)$$

$$\nabla_X f = \nabla_Y f * K^{\text{rot } 180}$$

(convolution of $\nabla_Y f$ and K after 180 degree rotation)

5. Consider a small perturbation δX in X

$$X + \delta X = (Q + \delta Q)^T (\Lambda + \delta \Lambda) (Q + \delta Q)$$

ignoring higher order terms:

$$X + \delta X = Q^T \Lambda Q + Q^T \delta \Lambda Q + (\delta Q)^T \Lambda Q + Q^T \Lambda (\delta Q)$$

removing X from both sides:

$$\delta X = Q^T \delta \Lambda Q + (\delta Q)^T \Lambda Q + Q^T \Lambda (\delta Q)$$

$$= Q^T \delta \Lambda Q + (Q^T \Lambda \delta Q)^T + Q^T \Lambda \delta Q$$

To simplify, let $A = \delta Q Q^T$, then,

$$\delta (Q Q^T) = \delta Q Q^T + Q \delta Q^T = 0.$$

$\Rightarrow A^T = -A$. ~~make~~ A is skew-symmetric

and $\delta Q = A Q$.

$$\text{Thus } \delta X = Q^T \delta \Lambda Q + Q^T \Lambda A Q - Q^T A \Lambda Q$$

$$= Q^T \delta \Lambda Q + Q^T (\Lambda A - A \Lambda) Q.$$

$$= Q^T \delta \Lambda Q + Q^T [\Lambda, A] Q$$

$$= Q^T \delta \Lambda Q + [A, \Lambda]$$

Gradient definition:

$$\delta f = \text{Tr}((\nabla_X f)^T \delta X)$$

$$= \text{Tr}((\nabla_X f)^T (Q^T \delta \Lambda Q + [A, \Lambda]))$$

$$= \text{Tr}(Q \nabla_X f Q^T \delta \Lambda) + \text{Tr}(\Lambda [A, \nabla_X f])$$

And we know

$$\delta f = \text{Tr}((\nabla_\Lambda f)^T \delta \Lambda) + \text{Tr}((Q^T \nabla_Q f Q)^T \Lambda)$$

Thus

$$Q \nabla_X f Q^T = \nabla_\Lambda f \Rightarrow \nabla_X f = Q^T \nabla_\Lambda f Q$$

$$(Q^T \nabla_Q f Q^*)^T = [\Lambda, \nabla_X f]$$

Given that $\nabla_x f$ is symmetric and Λ is diagonal, it can be simplified into:

$$[\Lambda, \nabla_x f]_{ij} = (\lambda_i - \lambda_j) (\nabla_x f)_{ij} \quad \text{for } i \neq j$$

Therefore:

$$(\nabla_x f)_{ij} = \frac{(\mathbf{Q}^T \nabla_x f \mathbf{Q})_{ij}}{\lambda_i - \lambda_j} \quad \text{for } i \neq j$$

this holds because for $i \neq j$, $\lambda_i \neq \lambda_j$.

Combining both parts:

$$\nabla_x f = \mathbf{Q} (\nabla_\Lambda f + \mathbf{M}) \mathbf{Q}^T$$

where

$$M_{ij} = \begin{cases} \frac{(\mathbf{Q}^T \nabla_x f \mathbf{Q})_{ij}}{\lambda_i - \lambda_j} & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases}$$

6. For i from 1 to n :
 For j from 1 to i :
 $s = X[i][j]$
 For k from 1 to $j-1$:
 $s = s - L[i][k] * L[j][k]$
 If $i == j$:
 $L[i][i] = \text{sqrt}(s)$
 Else:
 $L[i][j] = s / L[j][j]$

For all i, j :
 $L[i][j] = (1/c^2 \sum_{k=1}^j L[i][k] * L[j][k]) \neq \sum_{k=1}^j L[i][k] * L[j][k]$
 $\bar{x}[i][j] = 0$

For i from n down to 1:
 For j from i down to 1:
 $\bar{s} = 0$
 If $i == j$:
 $\bar{s} += (1/c^2 * L[i][i]) * L[i][i]$
 Else:
 $\bar{s} += (1/L[j][j]) * L[i][j]$
 $L[j][j] -= (s / (L[j][j]^2)) * L[i][j]$
 For k from $j-1$ down to 1:
 $\bar{s} -= L[i][k] * L[j][k] + L[j][k] * L[i][k]$
 $L[i][k] -= L[j][k] * \bar{s}$
 $L[j][k] -= L[i][k] * \bar{s}$
 $\bar{x}[i][j] += \bar{s}$